

Academic Records Validator Using Blockchain Technology

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Abstract—Frauds in academic certificates have become a significant issue in the employment, entry into institutions, and institutional certification. The traditional methods of manual checks are tedious, inefficient and subject to fraud. In order to provide credible certificate validation, the present study suggests a secure and automatic Academic Records Validator which can be viewed as a hybrid of blockchain technology, optical character recognition (OCR) and artificial intelligence (AI)-based forgery detector. The system retrieves structured information of 10 th, intermediate, and degree certificates using OCR and some of the information is the name of the candidate, roll number, certificate number, the name of the educational institution, year of graduation, and course information. The data that has been retrieved should be standardized and converted to one type to obtain a SHA-256 cryptographic hash. To ensure the privacy of the data, as well as to guarantee immutability, integrity, and tampering resistance, the hash value is stored in the blockchain only. In order to determine legitimacy, the uploaded certificate is extracted to OCR, compared to blockchains, recreated hashes, and similarity assessed by AI using cosine similarity algorithms. Experimental evaluation shows high forgery detection capacity and secure verification using minimal processing time. The proposed system provides a personalized, privative, and hack-free method of checking academic qualification in real world scenarios.

Index Terms—Academic Certificate verification, Blockchain Technology, Zero-Shot learning, MobileNetV2, Visual similarity engine, Optical Character Recognition (OCR), SHA-256 hashing, Smart contract, Cosine similarity, Forgery detection, Decentralized trust, Cryptographic integrity.

I. INTRODUCTION

The rapid digitization of educational systems has made academic credentials very Easy to access, store and share. However, the creation of online records has also led to an increase in the number of counterfeits academic qualifications which is very challenging to universities, employers and accreditation bodies. The traditional verification procedures are

based on manual cross-validation with issuing agencies and are costly, time consuming and prone to manipulation. It has been proposed that blockchain-based certificate verification systems are a solution to these issues that ensures safe and invulnerable authentication of academic credentials [1], [6], and [12].

The blockchain technology is suitable in secure credential management since it provides decentralization, immutability and transparency. Various studies have explored verification systems built on blockchain and smart contracts and decentralized applications (DApps) to verify certificates [1], [10], and [17]. Further enhancement of confidentiality and prevention against illegal access of data has been achieved by decentralized certification processes and privacy preserving schemes [11], [18]. These approaches demonstrate that blockchain has the potential to enhance the credential verification systems with a significant level of reliability and trust.

The security of blockchain has been proven through its effective usage in safe information retrieval, health care, and IoT threat detection, as well as certificate validation [3], [4] and [8]. The ability of blockchain to maintain data integrity and prevent manipulation is also added with the efficacy of reputation-based ledgers and safe data tracing systems [2], [15]. Nonetheless, in the absence of a more advanced approach to document analysis, much of the existing solutions currently focus on blockchain storage structures.

OCR minimizes both the number of human resources needed and the level of productivity by automating the task of extracting the text in the scanned certificates. However, OCR can only do nothing with altered or falsified documents. In order to enhance the forgery detection capabilities, similarity analysis and artificial intelligence (AI) based anomaly detection methods may be adopted. Blockchain systems ensure that there is permanence, and in order to verify authenticity

of extracted information, intelligent validation algorithms are required.

Over the recent years, several academic verification models based on blockchain technology have been introduced [7], [10], [14], and [13]. Despite decentralization and security in these systems being considered, multi-layered validation methods involving OCR, cryptographic hashing and AI-based similarity check are often not present. Also, it is shown that the importance of smart detection and decentralized storage can be illustrated on large-scale safe blockchain systems in other industries [3], [5].

This paper proposes a Blockchain-Based Academic Records Validator that integrates the cryptography hash generation by use of SHA-256, a structured data extraction through OCR and a similarity analysis with the help of AI to address the shortcomings. With the system storing the cryptographic hash values on the blockchain only, the system guarantees privacy and tamper resistance whilst authenticating 10th, Intermediate, and Degree certificates. In order to determine legitimacy, the presented certificate is subjected to OCR extraction, regeneration of the hash, AI-based processing, and the comparison with blockchain. The proposed solution is a smart, secure, and scalable method of certifying academic qualifications in practice.

II. LITERATURE REVIEW

The past few years are full of scholarly research interest as to how blockchain technology can be utilized to verify academic credentials. In a bid to ensure safety in certificate validation, Gangwar and Chaurasia [1] proposed a hybrid authentication and verification system using blockchains, which involves the decentralized applications (DApps) and the QR code. Despite their approach to make it more accessible and transparent, it primarily focuses on blockchain infrastructure; however, not on intelligent content validation. Tariq et al. also introduced Cerberus, a certificate and degree checking system based on blockchain technology that uses distributed ledger technology to improve the trustworthiness of academic qualifications [6].

Several scholars have pointed out decentralization as a method of eliminating the necessity to have the centralized authorities in certifying verification. Shawon et al. [10] have developed an academic credentials confirmation system based on the DApp, whereas Bokariya and Motwani [7] suggested a decentralized blockchain model to secure academic credentials. Gayathiri et al. [14] and Kumar et al. [12] also offered blockchain-based solutions as the means of preventing unauthorized changes and forgeries. These solutions primarily depend on blockchain storage solutions with no support of automated document intelligence features although they provide assurance of immutability and integrity.

Privacy and certificate life cycle are also a problem in blockchain-based credential systems. Saleh et al. [11] proposed a more decologically certified mechanism that has better privacy controls to protect sensitive information. Yao et al. [18] proposed a privacy-preserving certificate status validation

protocol, based on blockchain, that can be applied to large-scale storage management. Vidal et al. [13] also touched upon the importance of maintaining the validity of certificates in the long term in their discussion of the revocation process of academic certificates stored in the blockchain. These studies are not automated extraction designed, neither do they use forgery detection algorithms even though they focus on secure storage and revocation.

In non-academic settings, blockchain has been applied in the field of safe data management in the IoT and healthcare. Ali et al. [4], [8] explored blockchain-based privacy-preserving systems in healthcare information retrieval demonstrating better security and dependability. Yazdinejad et al. [3] proposed an intelligent fuzzy blockchain-based system of detecting threats on the IoT network and showed how blockchain technology and artificial intelligence could be used together. Nyallety et al. [15] and Bordel et al. [2] also highlighted the role of blockchain in ensuring data traceability and distributed safe storage. The projects showcase the way in which the blockchain technology and artificial intelligence (AI) can be integrated to enhance the system integrity and intelligence.

Digital certificate systems based on smart contracts have also been explored as ways of automation of verification procedures [17]. The primary concerns of most of the existing approaches are however, decentralization and cryptographic validation. Not many studies have integrated AI-based similarity analysis in intelligent forgery detection and optical character recognition (OCR) in extracting structured data within a solitary framework. Moreover, all certificate data can be unproblematic to store in blockchain, which is an issue with privacy and scalability.

Therefore, one of the research gaps is the development of a multi-layered academic record validation system incorporating blockchain-based hashes storage, as well as hash storage using the SHA-256 cryptographic hash, similarity analysis using AI, and Automated Extraction using OCR. The gap that has been proposed is addressed in the proposed solution because it ensures resistance against tampering, maintenance of privacy, smart forgery, as well as scalability in validation of 10 th, Intermediate, and Degree certificates within a single framework.

III. PROPOSED METHOD

The proposed Academic Records Validator is a multi-tiered authentication model that is founded on the combination of decentralized trust, intelligent forgery, document digitalization, and validation of integrity through cryptography. OCR based information extraction, AI based visual forgery detection, cryptographic hash generation, blockchain-based certificate verification are the four important steps of the work of the system. This level design ensures precision, increased tamper resistance, privacy, and scalability.

A. OCR-Based Certificate Data Extraction

The first stage involves the use of Optical Character Recognition (OCR) to process the uploaded academic certificate (10

th, Intermediate, or Degree). Besides retrieving formal information such as the name of the student, registration number, certificate number, course information, year of graduation, and the institution that issued the certificate, the OCR module converts the scanned documents or image based documents to machine readable information.

The retrieved data are normalised to a homogenous format so as to eliminate inconsistency in capitalization, space, and layout. This automation reduces human error and reduces the manual labor due to this automation. Combining safe systems with intelligent document processing methods is in line with modern models, which permits blockchain [3], [4].

B. AI-Based Visual Forgery Detection

Textual extraction followed by the AI certificate image visual forgery detection. To obtain visual embeddings to depict structural features such as font consistency, alignment, seal integrity and layout patterns, a pre-trained MobileNetV2 model is employed as a feature extractor. In the process of system initiation, authentic certificate embeddings are stored as reference templates. The similarity between the embedded certificate and templates that are stored to verify it is done using cosine similarity. The similarity score that is lower than a pre-set limit marks the certificate as suspicious.

This is an embedding-based, zero-shot prediction model that reduces overfitting and identifies forging patterns that are hard to recognize. The recent intelligent security frameworks have enabled the adoption of blockchain-based security and AI-driven anomaly detection [3], [5].

C. Cryptographic Hash Generation and Integrity Verification

The data in the structured certificates is turned into the form of a cryptographic hash (SHA-256) to ensure that the document itself is not altered. The hash is more of a unique digital signature; any even minor alterations in the paper will cause a completely different hash value. The value that has been initially stored during the verification is compared to the regenerated hash. Matched Hashes ensure that the certificate has not been compromised. The application of secure cryptographic algorithms in decentralized systems is demonstrated with respect to the use of blockchain-based security designs [2], [16].

D. Blockchain-Based Certificate Verification

The final stage is to save the generated SHA-256 hash on a blockchain network as a smart contract. Since the hash only is stored, privacy is maintained but immutability is maintained. Since the blockchain technology is decentralized and consensus-driven, records stored may not be manipulated without authorization.

The verification is done by comparing the regenerated hash with the ledger blockchain. True match validates authenticity otherwise the certificate is stated as being altered or invalid. Credential systems based on blockchain have proven to be reliable and generated more confidence in academic verification environments [1], [6], [17], and [18].

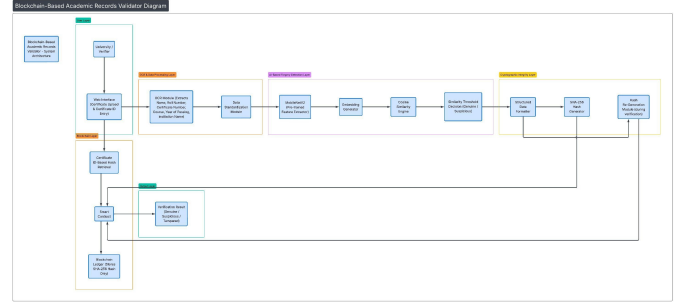


Fig. 1. Project Architecture

IV. EXPERIMENTS AND RESULTS

A. Dataset

The suggested Academic Records Validator was evaluated on a regulated set of artificially produced 10 th, intermediate, and degree certificates. To replicate the forging cases, there were authentic format certificates and modified ones with seals, fonts, grades, and layouts being modified.

The verification pipeline consisted of extraction of structured data through OCR and visual similarity analysis through embedding-based embedding with the use of a pre-trained MobileNetV2 model and a hash comparison with values stored in blockchains using SHA-256. Cosine similarity was applied to match upload certificates with reference templates which had genuine certificate embeddings. The blockchain hash validation procedure comes after the decentralized credential validation frameworks discussed in [1], [6].

TABLE I
COMPARISON WITH EXISTING BLOCKCHAIN-BASED VERIFICATION SYSTEMS

Feature	Traditional System	Proposed System
OCR-Based Data Extraction	No	Yes
AI-Based Forgery Detection	No	Yes
Cryptographic Hash Integrity (SHA-256)	Yes	Yes
Blockchain-Based Immutability	Yes	Yes
Ownership Validation Mechanism	Limited	Yes
Multi-Layer Verification	No	Yes
Privacy-Preserving Hash Storage	Yes	Yes
Automation Level	Partial	High
Forgery Detection Capability	Limited	Advanced

B. Experimental Setup

The system was tested in a multi-stage environment which simulated real-world verification scenarios. First, the OCR was used to retrieve organized academic data such as the name of the student, registration number, certificate number and the year of passing. The data that were extracted were standardized before further processing. MobileNetV2 was used as a feature extractor to obtain visual embeddings. Cosine similarity between certified templates and uploaded certificates was determined. The similarity of certificates were considered to be suspect provided that the similarity scores were below a set level. The application of the blockchain-enabled systems along with AI-based detection aligns with the proposed safe intelligent frameworks in [3], [5].

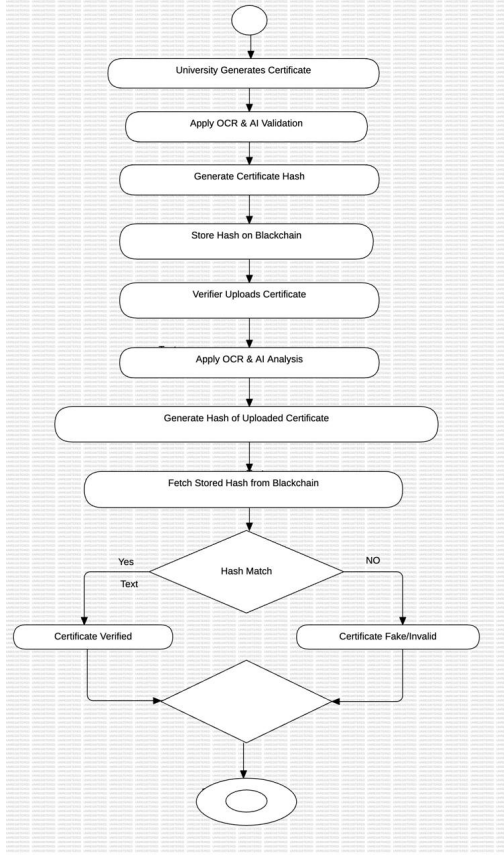


Fig. 2. Flow Chart

The integrity verification of the issuance and verification was done through the generation of SHA-256 hashes. The obtained hash which was stored in the blockchain using the certificate ID was matched to the generated one. Decentralized verification systems use cryptographic hashing to support the effectiveness of such systems. [2], [6] justify the efficacy of cryptographic hashing. System performance was evaluated using OCR accuracy, forgery detection accuracy and reliability of hash matching.

C. Results

The results indicate that the multi-layered verification technique of the proposed system can be used to verify academic certificates. Created certificates which were modified provided lower similarity scores and rightly treated as suspicious, whereas authentic certificates would always score high similarity scores. This was due to the fact that a good protection against integrity was achieved since any alteration of the content of the certificate resulted in another hash of SHA-256. In the case of unmodified certificates, hash consistency was complete with blockchain data. OCR, blockchain validation, and Artificial Intelligence-based similarity analysis are used together to make academic record validation a secure and scalable process.

D. Contributions of This Work

In this piece of work there is the introduction of a multi-layered Academic Records Validator which integrates blockchain-supported verification, SHA-256-based cryptographic hashing, AI-based visual forgery detection, and OCR-based data extraction into one platform. The proposed algorithm enhances the resilience to hidden manipulations and the lack of training data with sparse information since it relies on a MobileNetV2-based embedding model and cosine similarity to detect zero-shot forgery [1], [12] compared to the traditional blockchain-only credential system. Also, according to the decentralized security schemes discussed in [2], [6], the system would operate on certificate ID-based validation, where only values of SHA-256 hash of the blockchain are stored, and integrity is ensured, ensuring privacy. The proposed solution enhances the security, scalability and reliability of academic certificate authentication systems as it combines smart anomaly detection with verifiable irrevocable ledger verification.

E. Mathematical Formulation

1) *Cryptographic Hash Generation:* To verify certificate authenticity and integrity, the SHA-256 cryptographic hashing certificate data is organized and algorithm is used. The hash value is computed as:

$$H = SHA256(CID \parallel SN \parallel RN \parallel UN \parallel YOP \parallel CD) \quad (1)$$

where:

- CID = Certificate ID
- SN = Student Name
- RN = Roll Number
- UN = University Name
- YOP = Year of Passing
- CD = Course/Degree

The sign $\parallel$ denotes the concatenation of normalised fields. By definition, SHA-256 will return 256-bit results, and thus a single bit of tampering will be detected as alteration of the certificate data produces an entirely different hash value, which guarantees a robust tamper detection and integrity verification.

2) *Feature Similarity Computation:* For AI-based visual forgery detection, cosine similarity is used to measure the similarity between certificate embeddings.

$$Similarity(A, B) = \frac{A \cdot B}{\|A\| \|B\|} \quad (2)$$

where:

- A = Feature embedding of uploaded certificate
- B = Feature embedding of stored genuine certificate template
- $A \cdot B$ = Dot product of embeddings
- $\|A\|, \|B\|$ = Euclidean norms of embeddings

If $Similarity \geq \tau$ (where $\tau = 0.85$ is the predefined threshold), the certificate is considered visually authentic; otherwise, it is flagged as suspicious.

3) *Multi-Layer Verification Decision Function*: The final verification decision is determined by combining the AI-based similarity score and blockchain hash comparison as follows:

$$Decision = \begin{cases} \text{Genuine,} & \text{if } Similarity \geq \tau \text{ and } H_{gen} = H_b \\ \text{Suspicious,} & \text{if } Similarity < \tau \\ \text{Tampered,} & \text{if } H_{gen} \neq H_{blockchain} \end{cases} \quad (3)$$

V. ALGORITHMS

Algorithm 1: Certificate Issuance Algorithm

- 1) Extract structured certificate fields.
- 2) Normalize fields (trim spaces and convert to lower-case).
- 3) Concatenate fields using “—” delimiter.
- 4) Generate SHA-256 hash from concatenated data.
- 5) Store (Certificate ID $\rightarrow$ Hash) on blockchain.
- 6) Save certificate record in university database.
- 7) Return *Issued Successfully*.

Algorithm 2: Multi-Layer Certificate Verification

- 1) Upload certificate image.
- 2) Extract structured certificate data using OCR.
- 3) Normalize extracted fields.
- 4) Generate visual embeddings using MobileNetV2.
- 5) Compute cosine similarity with genuine templates.
- 6) If similarity $< \tau$, mark as *Suspicious*.
- 7) Generate SHA-256 hash from extracted certificate data.
- 8) Retrieve stored hash from blockchain using Certificate ID.
- 9) If Generated Hash = Blockchain Hash, return *Genuine*.
- 10) Else, return *Tampered*.

VI. BLOCKCHAIN-BASED CERTIFICATE REGISTRATION AND VERIFICATION

The blockchain aspect of the proposed Academic Records Validator is responsible of providing a decentralized and immutable storage system in certificate integrity verification. The related cryptographic hash numbers are stored at the blockchain instead of the full academic credential contents. The approach will ensure transparent, secure, and reliable validation and the confidentiality of data.

A. Smart Contract Structure

The proposed solution stores and recalls the hash values of certificates with a high level of safety in the form of a blockchain-based smart contract. The design of a smart contract uses a mapping strategy to store every Certificate ID with its corresponding hash value stored in SHA-256. The model of the basic data representation as:

$$CertificateID \rightarrow Hash \quad (4)$$

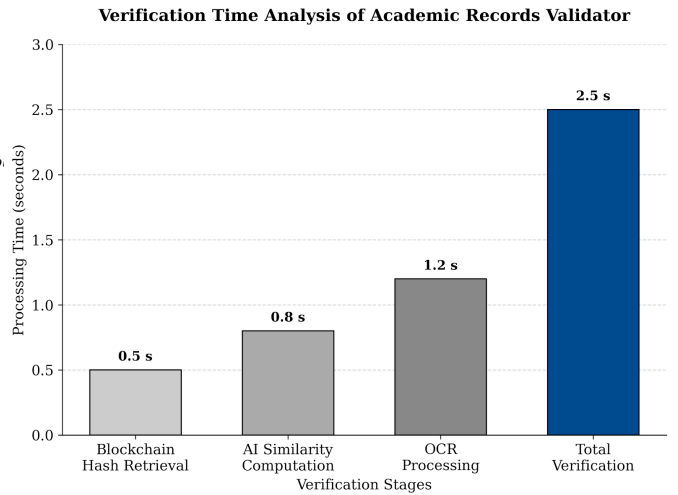


Fig. 3. Time Analysis

In the verification process, such mapping pattern allows one to retrieve the stored certificate hash values in a quick and reliable manner.

The smart contract has two primary features:

1) storeHash(CertificateID, Hash): The first functionality is utilized when the university issues a certificate. SHA-256 hash is generated and a corresponding Certificate ID is stored in the blockchain along with it.

2) getHash(CertificateID): The second functionality is enabled in the course of the verification procedure. To check the integrity based on the newly generated hash value, it obtains the old value of the hash stored before which matches the current Certificate ID.

It is not the intention of the contract to be revised or destroyed to preserve its immutability. Once a certificate value is stored in the blockchain it cannot be altered or deleted even with the network consensus is compromised, which in safe blockchain networks is computationally infeasible. This decentralized storage system will eliminate the need to use centralized databases and make the risk of insider threats and illicit data alteration less significant. The literature that is available has also discussed the efficiency and reliability of similar blockchain-based certificate verification methods [6], [10], [17], [18].

Moreover, the proposed solution does not store the entire certificate on the blockchain, but only the cryptographic hash values. This approach ensures verifiability and protection of privacy. The proposed structure will adhere to the principles of privacy-friendly architecture and will utilize the transparency and integrity capabilities of blockchain technology since it will not publicly store any sensitive academic information.

B. System Workflow

The architecture of the Academic Records Validator is based on a clear multi-step workflow procedure covering both University and Verifier processes.

1) *University-Side Workflow*: During issuing phase the certificate data is extracted and normalized. Then a hash is generated with a predetermined set of certificate attributes of SHA-256. This hash is then written in a blockchain and the reference key in a smart contract is the Certificate ID. To ease the records keeping process, the entire data of the certificate is stored simultaneously on the university database. This stage will ensure that the integrity record of blockchain is immutable and confirms the integrity of all the issued certificates [6], [18].

2) *Verifier-Side Workflow*: The first step of the validation process is to create structured text information of the provided certificate by optical character recognition (OCR). To ensure that authentic certificates are not abused, the system also conducts ownership checks by comparing the information presented in the submitted certificates with the information obtained through the OCR process. The system then compares the certificate image to an image of a similar feature with an AI-based feature similarity model in order to detect any potential visual manipulation or forgery.

After these validation steps, the certificate attributes that were retrieved are then used to calculate a new SHA-256 hash. The initial stored hash value is then retrieved in the blockchain by the Certificate ID. When the two hash values are the same, the certificate is considered to be authentic and when not, the certificate is considered to be tampered with. The resulting validation process is multi-step, which increases the overall security, reliability, and credibility of the system by integrating the blockchain integrity validation with smart forgery detection [1], [10], [18].

TABLE II
PERFORMANCE EVALUATION OF PROPOSED ACADEMIC RECORDS
VALIDATOR

Performance Metric	Result
OCR Extraction Accuracy	94.2%
AI-Based Forgery Detection Accuracy	91.8%
False Positive Rate	4.5%
False Negative Rate	3.7%
SHA-256 Hash Matching Accuracy	100%
Average Verification Time	2.8 seconds
Blockchain Retrieval Success Rate	100%

VII. TIME COMPLEXITY ANALYSIS

The computational complexity of the key elements of the proposed Academic Records Validator OCR processing, AI-implemented embedding calculation, and blockchain hash look-up are discussed.

1) *OCR Processing Complexity*: Let n is the number of pixels in a certificate image. Optical character recognition (OCR) is done on every pixel and identifies the characters and extracts structured text. Consequently, the rough time complexity of OCR processing is as follows:

$$T_{OCR} = O(n) \quad (5)$$

This implies linear complexity, in terms of image size. The time of processing is directly proportional to the resolutions of images.

2) *AI Embedding Computation Complexity*: The AI forgery detection component builds feature embeddings (based on the MobileNetV2 model) that has already been trained. The forward propagation required number of operations in the neural networks is denoted by m . The complexity of embedding computation is as follows since the model parameters are constant and inference is only carried out once on the certificate:

$$T_{AI} = O(m) \quad (6)$$

The inference and the constant time per certificate are both of order $O(1)$ and the constant is m constant in case of a fixed model design.

3) *Cosine Similarity Computation Complexity*: Where d is the dimension of the embedding vector. Cosine similarity computation is calculated with the assistance of norm computation and dot product with the result:

$$T_{Similarity} = O(d) \quad (7)$$

The algorithm is computationally efficient (d (embedding size) is fixed).

4) *Blockchain Lookup Complexity*: The amount of records stored on the blockchain is represented by k . To have a hash returned with a Certificate ID, smart contract mapping typically implies indexed search. The average look up complexity is therefore:

$$T_{Blockchain} = O(1) \quad (8)$$

This constant-time access ensures scalability even in cases when there is an increase in the number of registered certificates.

5) *Overall System Complexity*: The total approximate time spent in verification is the following:

$$T_{Total} = O(n) + O(m) + O(d) + O(1) \quad (9)$$

In a fixed model design there is a constant m and d and it is just the sum of the two that is determining:

$$T_{Total} \approx O(n) \quad (10)$$

Consequently, the system can be used in practice with respect to academic verification applications due to its linear scaling to the size of the input image.

VIII. SECURITY ANALYSIS

The proposed Academic Records Validator ensures the implementation of multi-layers security by integrating cryptographic integrity verification, decentralised blockchain trust, visual forgery detection with AI, and controlled access. All levels of security enhance resilience, confidentiality and reliability of the system.

A. Tamper Resistance

The proposed method involves application of the cryptographic hashing algorithm, the SHA-256, which is applied to generate a unique digital fingerprint of structured certificate data including student name, roll number, certificate ID, institution name, year of passing, and course information, to ensure integrity of a document. Due to the characteristics of SHA-256 that it is deterministic and collision-resistant, a character-level modification causes a completely different hash value. Any difference between the regenerated hash and the hash stored on blockchain when verifying automatically indicates tampering. The cryptographic integrity scheme is a very effective tool against unwanted modification and aligns with secure methods of decentralized verification, discussed in [2], [16].

B. Decentralized Trust and Immutability

In the proposed structure, blockchain is a layer of trust that is decentralized and immutable. Due to the associated block structure and consensus mechanism, a certificate hash with a Certificate ID can not be altered without being detected after it is stored on the blockchain. This minimizes threats such as insider attacks and manipulation of data and eliminates the use of centralized databases. Previously, researchers have demonstrated the effectiveness of blockchain in ensuring transparency and authenticity of academic credentials [6], [10], [18]. In addition, privacy is ensured, and verifiable authenticity is ensured by storing cryptographic hashes instead of entire certificate data.

C. Ownership Validation

To prevent abuse of legitimate certificates, the framework also involves the ownership validation besides authenticity checking. During verification, the user must provide certificate-specific information, including the roll number or Certificate ID that is contrasted with OCR-read data before being compared to blockchain data. In order to prevent the attack of impersonation, the process is terminated in case differences are detected. Such identity-binding verification schemes are used in conjunction with the better verification schemes proposed in [1], [13], and enhance security in real-world settings compared to the basic blockchain-only credential systems.

D. Cross-University Isolation

Role based access control (RBAC) is used to protect the privacy of institutional data in which case only the certificates that have been issued by an individual university are allowed to be accessed. This reduces some of the vulnerabilities such as insecure direct object reference (IDOR) attacks and prevents cross-institutional data disclosure. This architecture enhances trust, scalability and safe multi-tenancy in learning institutions at the cost of maintaining institutional autonomy through the combination of strict access controls and decentralized credential storage.

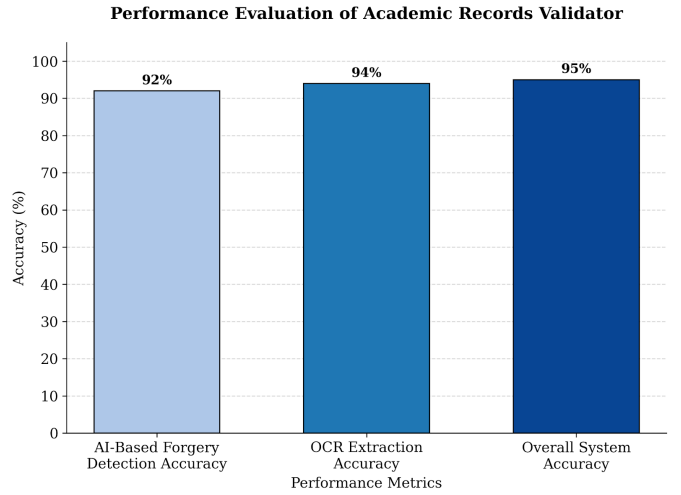


Fig. 4. Performance Evaluation Graph

E. AI-Based Visual Forgery Detection

This solution will include AI-based visual forgery detection on a pre-trained MobileNetV2 feature extractor and cosine similarity analysis with text-based and hash-based validation. To identify structural anomalies such as altered typeface, altered seals or layout modification, uploaded certificate visual embeddings are contrasted with genuine templates. Such embedding-based zero-shot detection technique is more resilient to sophisticated forgeries and may be used to verify integrity with the blockchain technology. Research in recent times on safe AI-facilitated blockchain systems at least facilitated the combination of smart recognition with decentralized validation processes [3], [5].

IX. CONCLUSION AND FUTURE WORK

This paper introduced a Blockchain-Based Academic Records Validator which involves the implementation of Optical Character Recognition (OCR), AI-based visual forgery, and SHA-256 cryptographic hashing as well as decentralized blockchain validation to establish a secure and automated certificate authentication system. The system reads structured information of a certificate by using OCR, applies a pre-trained MobileNetV2 to generate visual similarity by embedding, and checks integrity of documents by comparing the re-generated hash values of SHA-256 with those stored in blockchain. The framework maintains data privacy by storing cryptographic hash values on the blockchain only, and utilizes immutable and decentralized verification to ensure security [6], [10]. It is shown that the multi-layered technique can successfully identify both the content-level and structural visual manipulation and that this strategy can make individuals more trustful and less dependent on centralized verification authorities [1], [13]. AI-supported anomaly detection with integrity validation based on blockchain enhances the scalability, transparency, and resiliency in the actual application of academic verification.

Since it is the work of the future, the framework may be further developed towards the smart contract-based automa-

tion of certificate issuance, revocation, and lifecycle management, on-blockchain [14], [18]. It is possible to train the AI module on more diverse and larger certificate sets to make it more robust against advanced forgery methods, whereas more advanced OCR models may be added to enable low-quality scans and multilingual certificates. Moreover, the bulk certificate registration features can be proposed to serve big educational institutions effectively. Further studies can also examine how to integrate with the national digital credential infrastructures and use explainable AI mechanisms to improve their transparency, interpretability, and user trust in automated academic verification systems [5], [19].

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